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A

B.Tech. Degree VI Semester Examination April 2018

ME 15-1605 CAD/CAM
(2015 Scheme)

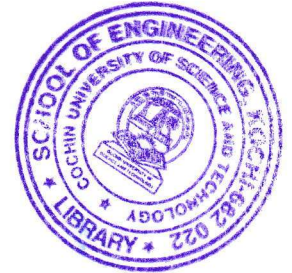
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Explain any four geometric transformations used in CAD with appropriate transformation matrices.
- (b) What is computer aided process planning (CAPP)? What are the challenges in developing generative type CAPP?
- (c) List out and explain different levels of automation.
- (d) What is line balancing? What is balance delay?
- (e) Explain the terms (i) NC (ii) CNC (iii) DNC and (iv) FMC.
- (f) What are the different types of statements used in APT? Explain each with an example.
- (g) Explain the different static and dynamic tests performed on CNC machines.
- (h) Explain the types and features of spindle bearings for machine tools.
- (i) Explain the structure of a SCARA robot with the help of a neat figure.
- (j) Compare textual language programming with teaching method in robotics.



PART B

(4 × 10 = 40)

- II. (a) Explain (i) extrusion (ii) revolve (iii) loft and (iv) sweep functions in solid modeling with the help of neat figures. (8)
- (b) Explain constructive solid geometry with the help of an example. (2)

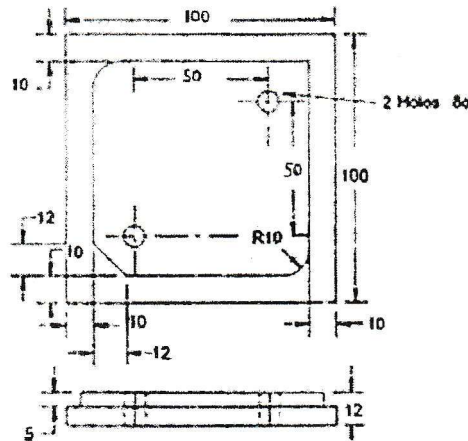
OR

- III. Draw the network diagram and solve the following assembly line balancing problem using Largest Candidate Rule. The production rate required is one assembly every 15 min. Find balance delay. (10)

Task	Time	Immediate Predecessors
A	12	Nil
B	6	A
C	6	B
D	2	B
E	2	B
F	12	B
G	7	C, D
H	5	G
I	1	E
J	4	I, F
K	6	H, J
L	7	K

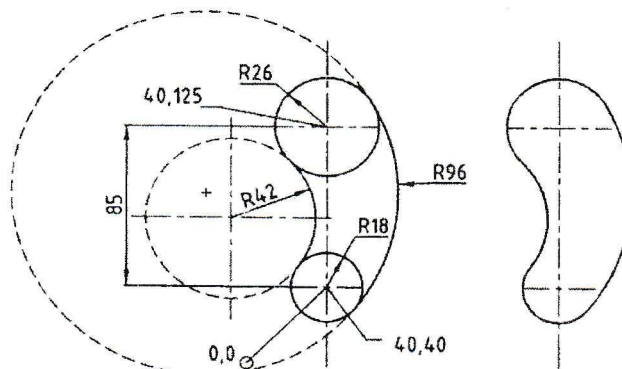
(P.T.O.)

- IV. A component to be machined is shown in the figure below. The outer profile (step 5 mm deep) needs to be machined using an end mill of diameter 16 mm (HSS). Prepare the NC code. (10)
(The work piece provided is 100 mm × 100 mm × 12 mm)

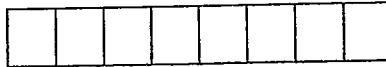


OR

- V. A component as shown in the figure below is to be machined from an MS sheet of 5 mm thickness. Write APT code for milling the profile with a $\phi 10$ mm end mill. (10)



- VI. Explain the working principles of an optical rotary encoder for CNC machines with the help of neat figures. A ball screw has a pitch of 10 mm. How many radial gratings are required for an accuracy of 1 μ m if two optical sensors are used at 90° phase difference? (10)
- OR
- VII. What are modular tools? List out and explain the components of modular tools with neat figures. Explain the cost advantage in using modular tools compared to custom made tools. (10)
- VIII. Explain the criteria in the selection of Robotic configurations programming techniques, payload and accuracy for their applications in (i) assembly (ii) arc welding. (10)
- OR
- IX. Explain with appropriate illustrations (i) hydraulic and pneumatic components, valves, actuators and their control (ii) AC and DC servo motors and their control (iii) various sensors used in electric, hydraulic and pneumatic systems, and sensors used in robotics such as vision systems, range detectors, force and torque sensors. (10)



B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

**ME 19-205-0605 CAD/CAM
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of modelling and automation.
 CO2: Study the basics of computer numerical control.
 CO3: Study the design features of CNC machines and RP processes.
 CO4: Understand the basic concepts of robotics.

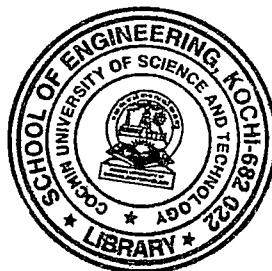
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

**PART A
(Answer *ALL* questions)**

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Differentiate between constructive solid geometry (CSG) and boundary representation method (BREP).	3		L2	1	1
(b) What is the production rate and throughput time required for each cycle of an 8-station dial indexing automatic assembly machine powered by a Geneva mechanism operating at a driver rotation speed of 30 revolutions per minute.	3		L3	1	1
(c) Differentiate between CNC and DNC with neat schematic diagrams.	3		L2	2	1
(d) What are the different types of statements used in APT? Explain each with an example.	3		L1	2	1
(e) Explain forward kinematics and inverse kinematics with respect to robot manipulation.	3		L1	4	1
(f) Compare the characteristics of PTP (Point-to-Point) and CP (Continuous Path) robots and assign each type of robot to the appropriate applications of loading and unloading, spot welding, arc welding and spray painting.	3		L1	4	1
(g) Explain the basic principles of Rapid Prototyping.	3		L1	3	1
(h) What is post processing in additive manufacturing? Why post processing is required?	3		L2	3	1

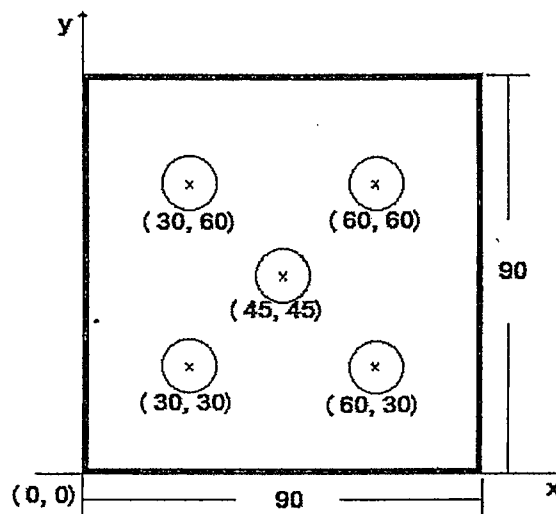
(P.T.O.)



PART B

(4 × 12 = 48)

- II. A 10 station transfer machine is under consideration to produce a component used in the blower. The item is produced by more conventional methods, but demand for the item cannot be met. The manufacturing engineering department has estimated that the ideal cycle time will be $T_c = 1$ min. from similar transfer lines, it is estimated that breakdowns of all types will occur with a frequency of $F = 0.10$ breakdown/cycle and that the average downtime per line stop will be 6 min. The casting I for the component costs ₹114.15 and it will cost ₹76.10 per minute to operate the transfer line. Cutting tools are estimated to cost ₹11.42 per work part. Compute the following measures of line performance
- Production rate
 - Line efficiency
 - Cost per unit produced.
- OR**
- III. (a) An automatic assembly machine with 8 stations, driven by a Geneva mechanism, has an indexing time of 3 seconds and a dwell time of 5 seconds. Under ideal conditions, the machine completes one assembly in 8 seconds, resulting in a production rate of 450 units per hour. Each station malfunctions once every 100 cycles and if any malfunction occurs, the machine jams, requiring an operator to adjust and restart it, which takes 10 minutes. Calculate the following:
- The percentage of downtime
 - The new production rate
 - The line efficiency.
- (b) Calculate the new transformation matrix for a triangle defined by the coordinates A(2,6), B(8,10) and C(10,3), after applying a translation of 4 units in the positive x-direction and 3 units in the positive y-direction.
- IV. (a) What is the difference between absolute and incremental programming? 3 L1 2 1
- (b) What are G codes and M codes in CNC programming? 3 L2 2 1
- (c) Write a CNC part program to drill 5 holes (R 3 mm) as shown in the figure at an aluminium slab of 19 mm thickness and size 90 mm × 90 mm. Use G81 fixed cycle for drilling. 6 L3 2 2



(All dimensions in mm)

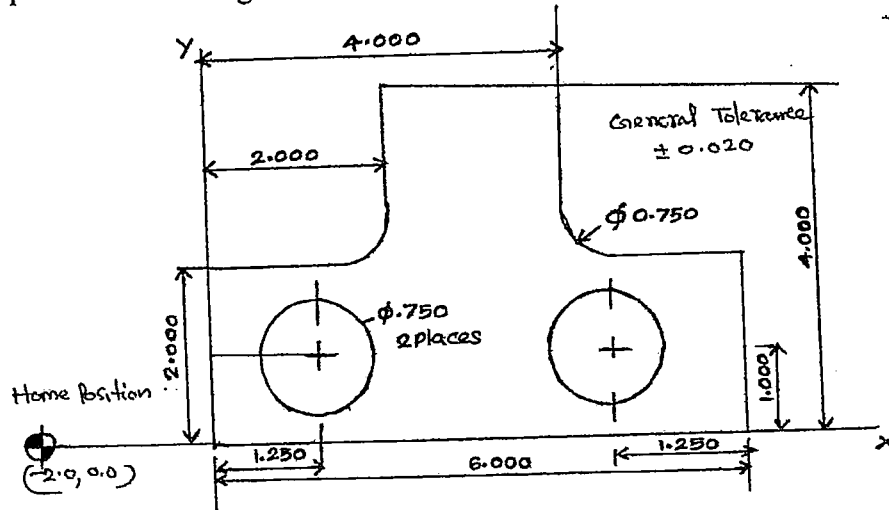
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BTS-VI(R/S)-05-24-2461

	Marks	BL	CO	PO
V.	12	L3	2	2

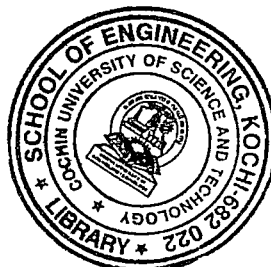
Figure shows a two and a half dimensional part. The part thickness is 1 inch. Two holes in the positions shown are to be drilled in the part. Write an APT program to drill the two holes. Use the tool home position shown a drilling speed of 500 r/min. The machine that performs the drilling is number5 and its controller is coded as DRILL.



VI. (a)	Describe the control loops used in robotic systems. Explain the role of feedback in maintaining the accuracy and stability of robotic movements.	6	L2	4	1
(b)	Compare different types of sensors, such as vision systems, range detectors and force sensors. Explain how these sensors enhance robot perception and interaction with the environment.	6	L2	4	1
OR					
VII. (a)	Draw a neat diagram of articulated robot with 5 joints.	4	L2	4	1
(b)	Compare and contrast different programming methods used in robotics, such as manual teaching and lead-through teaching and explain the advantages and challenges associated with each method.	8	L2	4	1
OR					
VIII. (a)	Explain the process chain in RP in Integrated CAD/CAM environment.	8	L2	3	1
(b)	What is the working principle of stereolithography in Rapid Prototyping?	4	L2	3	1
OR					
IX. (a)	Explain DNC, CAE, CIM, FMS and basic concepts of AI and expert systems for manufacturing automation.	8	L2	3	1
(b)	Explain the working principle of Fused Deposit Modelling RP process.	4	L2	3	1

Blooms's Taxonomy Level's

L1 – 34.78%, L2 – 39.13%, L3 – 26.08%.



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B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0605 CAD/CAM
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of Computer Aided Design (CAD), geometric modelling, automation, flow line analysis, and line balancing.
- CO2: Understand the structure and special design features of CNC machines. Learn manual CNC programming and computer-aided programming technique like APT.
- CO3: Understand the basic configurations, components, applications and the programming of robots.
- CO4: Learn the fundamentals of the advanced concepts in automation and manufacturing like DNC, CIM, FMS, and CAE and the basic concepts of AI and expert systems for manufacturing automation. Learn the basic concepts and applications of Rapid Prototyping.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

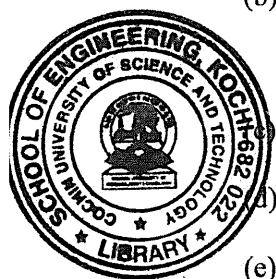
(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Explain: (i) orthographic projection (ii) isometric projection (iii) axonometric projection with illustrations.	3	L2	1	1,2,5
(b) Define and explain the terms with appropriate mathematical expressions: (i) manufacturing lead time (ii) production capacity (iii) production rate (iv) cycle time (v) availability (vi) work in progress.	3	L2	1	1,2,5
Describe the axes' nomenclature of a CNC Turning Centre and CNC Turning Centre with an illustration.	3	L3	2	1,5
(d) What are zero offsets? Explain how zero offset is determined in a CNC lathe.	3	L2	2	1,2,3
(e) Explain the difference between point-to-point and contour type motion systems in robots	3	L2	3	1,2,4
(f) Explain any three examples of end effectors used in robot with neat figures.	3	L2	3	1,2,4
(g) Describe the limitations of rapid prototyping methods.	3	L1	4	1,4,5
(h) List out and explain different levels of automation with illustrations.	3	L1	4	1,2

PART B

(4 × 12 = 48)

- II. (a) What are transformation matrices? Give transformation matrices for: (i) Translation (ii) Rotation (iii) Reflection in homogeneous coordinates in 2D. 6 L1 1 1,2,5
- (b) The coordinates of the vertices of a rectangle are (0, 0), (2, 0), (2, 2), and (0, 2). Construct a matrix with homogeneous coordinates of the rectangle and rotate it by 45 degrees about the centre (1, 1). Use appropriate transformation matrices for operation. 6 L3 1 1,2,5

OR**(P.T.O.)**

BTS-VI(R/S)-06-23-1822

- III. (a) (i) What are the different plant layouts? Explain with illustrations. Marks 2 BL L1 CO 1 PO 1,4,5
(ii) What is the line balancing problem? What is balance delay? 2
(b) In a factory 480 units are to be made in an 8 h (480 min) shift. The activity times and precedence requirements are shown in the table given below. Assuming 100% efficiency, solve the line-balancing problem using the largest candidate rule and find the balance delay. 8 L3 1 1,2,4,5

Task	Task Time (Seconds)	Precedence
A	15	-
B	23	A
C	17	B
D	42	B
E	15	B
F	37	C
G	5	D, E
H	12	F, G
I	34	H
J	27	H
K	18	I, J
L	7	K

- IV. (a) Explain the classification of CNC machines based on control loop used with illustrations. 4 L1 2 1,2,5
(b) A CNC program for milling a profile is given below. Draw the cutter path and the profile of the job. The thickness of the plate is 5 mm. An HSS endmill, $\phi 20$ mm is used for milling the profile. 8 L3 2 1,2,5
 $d=20$ mm;
T1- end mill $\phi 20$ mm

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%1234
N1 G54 G90 G94
N2 T1 M06
N3 M03 S330 M08
N4 G00 X-15 Y-15 Z5 (5 mm above the job surface)
N5 G01 Z-10 F100 (move to the required depth)
N6 X-10
N7 Y80
N8 G02 X30 Y110 R40
N9 G01 X74.14
N10 X130 Y54.14
N11 Y-10
N12 X-15
N13 G00 Z100
N14 G28
N15 M02

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OR

- V. (a) Describe Recirculating Ball Screws used in CNC machines. How are Ball Screws pre-loaded? How does preloading affect the accuracy of Ball Screws? 6 L2, L3, L5 2 1,2,3,5
(b) (i) What is machine zero? Illustrate machine zero for CNC Turning Centres and Machining Centres. 6 L3 2 1,2,5
(ii) What is referencing (Homing) in CNC Machines?

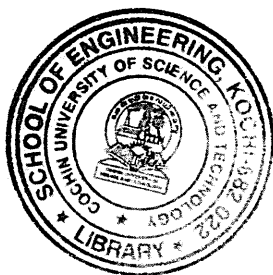
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BTS-VI(R/S)-06-23-1822

		Marks	BL	CO	PO
VI.	(a) Explain the structure of a typical robot with neat figure.	6	L2	3	1,2,3
	(b) Explain following kinematic configurations, applications, and its work envelope with neat figure:	6	L2	3	1,2,3
	(i) Cartesian manipulator				
	(ii) Spherical manipulator.				
OR					
VII.	Make a conceptual design of a simple pick and place robot which pick material from one conveyor belt and place it in another conveyor belt. Explain the configuration of the system and the components with the help of neat diagrams.	12	L6	3	1,2,3,4
VIII.	(a) Explain stereolithography with the help of neat figure.	6	L2	4	1,2
	(b) Explain the organizational structure for CIM. Which functional groups come under CIM?	6	L3	4	1,4
OR					
IX.	(a) What are the steps involved in a typical material addition rapid prototyping?	6	L2	4	1,2
	(b) Explain FMS with illustration. Differentiate between FMC and FMS.	6	L2	4	1,2

Bloom's Taxonomy Levels

L1 = 16.7%, L2 = 39.2%, L3 = 32.5%, L5 = 1.6%, L6 = 10%.



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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0605 CAD/CAM
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

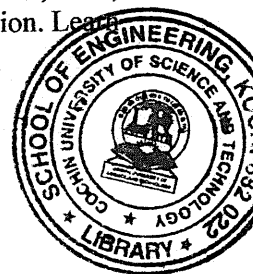
Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of Computer Aided Design (CAD), geometric modelling, automation, flow line analysis, and line balancing
- CO2: Understand the structure and special design features of CNC machines. Learn manual CNC programming and computer-aided programming technique like APT.
- CO3: Understand the basic configurations, components, applications and the programming of robots.
- CO4: Learn the fundamentals of the advanced concepts in automation and manufacturing like DNC, CIM, FMS, and CAE and the basic concepts of AI and expert systems for manufacturing automation. Learn the basic concepts and applications of Rapid Prototyping.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A

(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.						
(a)	List down the steps in the general design process (Shigley's design process) and explain the use of computers in each step. Indicate the name of a CAD package used in each step.		3	L2	1	1,2,5
(b)	Explain flexible automation. Discuss the type of industry where flexible automation is most desirable.		3	L3	1	1,2,5
(c)	What are the components of a CNC machine? Explain with proper illustration.		3	L1	2	1,5
(d)	What is the stick-slip phenomenon in machine tool guideways? How can the stick-slip be eliminated in guideways?		3	L2	2	1,2,3
(e)	What are the features of the robots used for the assembly operation? Suggest the best robotic configuration for assembly with justification.		3	L4	3	1,2,4
(f)	Explain the textual language programming of robots. List down and explain any four AML commands.		3	L1	3	1,2,5
(g)	Write short notes on CIM, FMS and DNC.		3	L1	4	1,4,5
(h)	Explain the challenges in the additive manufacturing of metals.		3	L5	4	1,2,5,6

PART B

(4 × 12 = 48)

II.	(a)	Explain the following techniques in solid modeling:	6	L1	1	1,2,5
	(i)	Extrusion				
	(ii)	Revolution				
	(iii)	Lofting				
	(iv)	Sweeping.				
	(b)	Rotate the square with coordinates (1,1), (3,1), (3,3) and (1,3) about the point (2, 2). Use homogeneous coordinates and concatenation of translation and rotation matrices.	6	L4	1	1,2,5

OR

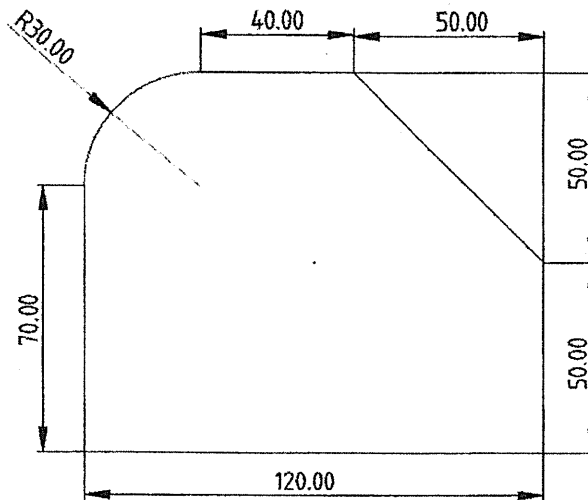
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BTS-VI(R)-04.22-0310

- III. (a) What are the different levels of automation? Explain each with a proper illustration. 4 L1 1 1,4,5
- (b) A new small electrical appliance is to be assembled on a production flow line. The total job of assembling the product has been divided into minimum rational work elements. The table shows the elemental times and their immediate predecessors. The cycle time is 1 minute and the line efficiency is 100%.
- (i) Construct the precedence diagram from the data provided on work elements
- (ii) Use the largest-candidate rule to assign the work elements to stations. Find out the balance delay.

Activity No	1	2	3	4	5	6	7	8	9	10	11	12
Element Time T_{ej}	0.2	0.4	0.7	0.1	0.3	0.11	0.32	0.6	0.27	0.38	0.5	0.12
Must be preceded by	-	-	1	1,2	2	3	3	3,4	6,7,8	5,8	9,10	11

- IV. (a) How are NC/CNC machines classified? Explain each type with appropriate illustrations. 4 L1 2 1,2,5
- (b) Write a CNC program to mill the profile shown in the figure from a plate of appropriate size. The thickness of the plate is 5 mm. Use an HSS endmill, $\phi 20$ mm for milling the profile. 8 L3 2 1,2,5



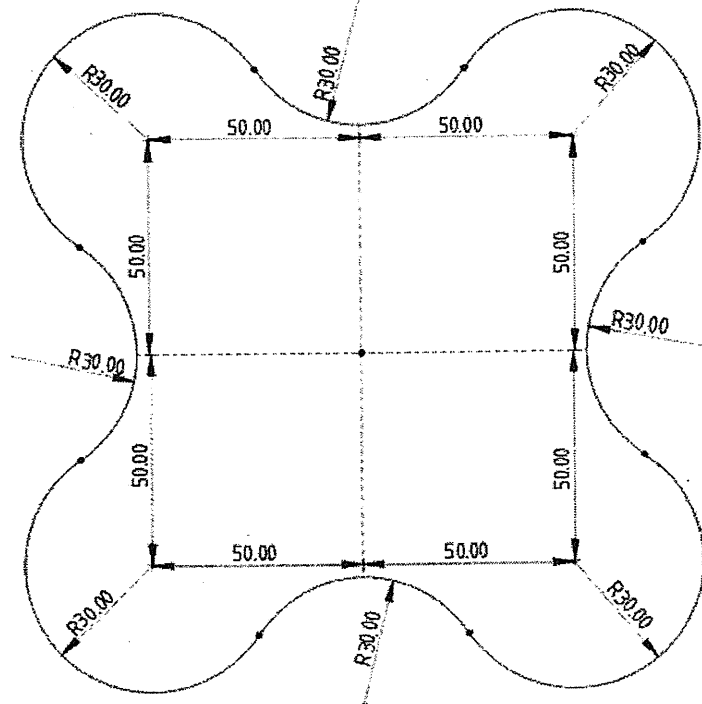
All dimensions in mm

OR

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BTS-VI(R)-04.22-0310

- V. (a) Compare ball screws with lead screws in terms of efficiency, performance and other features. How is pre-loading of ball screws done? 4 L5 2 1,2,3,5
- (b) Write an APT programme for generating CNC code for the profile given below from a mild steel plate of $170 \times 170 \times 5$ mm size. Use a carbide endmill of 20 mm diameter. All dimensions are in mm. 8 L3 2 1,2,5



- VI. (a) Draw the work envelope of
(i) a SCARA
(ii) an articulated robot. 6 L2 3 1,2,3
- (b) Explain in brief the control loops used in robotic systems. 6 L1 3 1,2,3,5
- OR**
- VII. Make a conceptual design of a gantry robot (rectangular/cartesian configuration) for loading and unloading a CNC machining centre. Explain the configuration of the system and the components with the help of neat diagrams. 12 L6 3 1,2,3,5
- VIII. (a) What is adaptive (intelligent) manufacturing? Evaluate different strategies used in adaptive manufacturing. 6 L5 4 1,2,5
- (b) Discuss the use of expert systems in manufacturing. 6 L3 4 1,4
- OR**
- IX. (a) Explain selective laser sintering (SLS) with the help of neat figures. 6 L2 4 1,5
- (b) Explain selective laser melting (SLM) with the help of neat figures. 6 L2 4 1,5

L1-30% L2-22% L3-18% L4=13% L5-13% L6-4%
